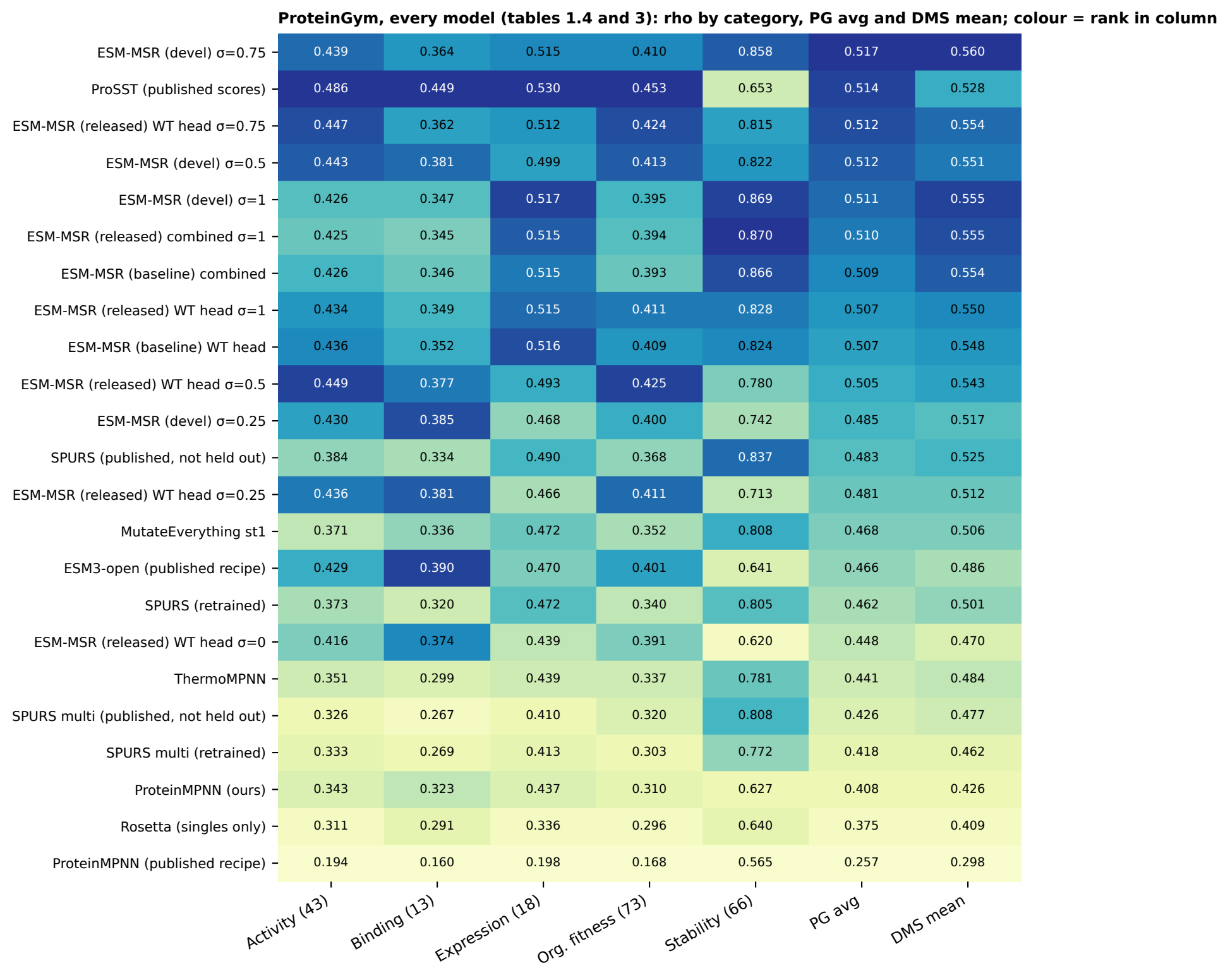
ProteinGym, every model (tables 1.4 and 3): rho by category, PG avg and DMS mean; colour = rank in column
ESM-MSR (devel) σ=0.75 0.439 0.364 0.515 0.410 0.858 0.517 0.560
ProSST (published scores) 0.486 0.449 0.530 0.453 0.653 0.514 0.528
ESM-MSR (released) WT head σ=0.75 0.447 0.362 0.512 0.424 0.815 0.512 0.554
ESM-MSR (devel) σ=0.5 0.443 0.381 0.499 0.413 0.822 0.512 0.551
ESM-MSR (devel) σ=1 0.426 0.347 0.517 0.395 0.869 0.511 0.555
ESM-MSR (released) combined σ=1 0.425 0.345 0.515 0.394 0.870 0.510 0.555
ESM-MSR (baseline) combined 0.426 0.346 0.515 0.393 0.866 0.509 0.554
ESM-MSR (released) WT head σ=1 0.434 0.349 0.515 0.411 0.828 0.507 0.550
ESM-MSR (baseline) WT head 0.436 0.352 0.516 0.409 0.824 0.507 0.548
ESM-MSR (released) WT head σ=0.5 0.449 0.377 0.493 0.425 0.780 0.505 0.543
ESM-MSR (devel) σ=0.25 0.430 0.385 0.468 0.400 0.742 0.485 0.517
SPURS (published, not held out) 0.384 0.334 0.490 0.368 0.837 0.483 0.525
ESM-MSR (released) WT head σ=0.25 0.436 0.381 0.466 0.411 0.713 0.481 0.512
MutateEverything st1 0.371 0.336 0.472 0.352 0.808 0.468 0.506
ESM3-open (published recipe) 0.429 0.390 0.470 0.401 0.641 0.466 0.486
SPURS (retrained) 0.373 0.320 0.472 0.340 0.805 0.462 0.501
ESM-MSR (released) WT head σ=0 0.416 0.374 0.439 0.391 0.620 0.448 0.470
ThermoMPNN 0.351 0.299 0.439 0.337 0.781 0.441 0.484
SPURS multi (published, not held out) 0.326 0.267 0.410 0.320 0.808 0.426 0.477
SPURS multi (retrained) 0.333 0.269 0.413 0.303 0.772 0.418 0.462
ProteinMPNN (ours) 0.343 0.323 0.437 0.310 0.627 0.408 0.426
Rosetta (singles only) 0.311 0.291 0.336 0.296 0.640 0.375 0.409
ProteinMPNN (published recipe) 0.194 0.160 0.198 0.168 0.565 0.257 0.298
Activity (43)
Binding (13)
Expression (18)
Org. fitness (73)
Stability (66)
PG avg
DMS mean